

Python in Electrical Engineering

Database Operations with Pandas

This set of exercises focuses on **data processing using SQLite databases and Pandas**. Tasks include querying relational data, merging tables, performing aggregations, and analyzing sales/customer metrics from the *Northwind* sample database (a simulated trading company dataset).

Task 1

- A sample SQLite database is provided, containing data related to commercial activity, including order management, customers, suppliers, and products (<https://github.com/jpwhite3/northwind-SQLite3>).
- Establish a connection to the database and familiarize yourself with its structure.

Task 2

Load the **Customers** and **Orders** tables and perform the following operations in Pandas:

- Display a list of unique cities (**City**) where customers are located.
- Count how many customers come from each city.
- Find the **5 customers** who placed the most orders (number of orders per **CustomerID**).

Task 3

Load the **Order Details** and **Products** tables and perform the following operations in Pandas:

1. Merge the data to obtain order details along with product names.
2. Count how many times each product was ordered (sum of **Quantity**).
3. Find the product that generated the highest total revenue (**UnitPrice * Quantity**).

Task 4

Load the **Orders**, **Order Details**, **Products**, and **Customers** tables and perform the following operations in Pandas:

1. Merge the data from these tables to obtain order details along with customer and product names.
2. Calculate the **total value** of all orders placed by customers from each country (**Country**).
3. Find the customer who generated the **highest revenue**.